

Locally Robust Commensurate BART for Borrowing External Controls

Local compatibility and borrowing calibration

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1. Established borrowing priors use one study-level parameter. Compatibility with an external control can vary across patient profiles.
2. LRC-BART adds a sparse discrepancy ensemble to the pooled BART surface, with a spike-and-slab commensurate prior on each discrepancy leaf.
3. We calibrate a variance-based prior ESS for the standardized control mean. Its ceiling depends on the RWD fit at trial profiles.
4. Simulations with regional shifts, and a 30-patient single-arm myeloma trial with a registry control.

Potential gain

- More control observations
- Greater precision
- Smaller concurrent control arms

Why pooling can mislead

- Different eligibility and care pathways
- Different covariate distributions
- Measured and unmeasured confounding

External controls may agree with the trial in one patient group and differ in another.
A single borrowing parameter cannot distinguish these groups.

Established priors use a study-level borrowing parameter

Meta-analytic predictive prior (Neuenschwander et al., 2010; Schmidli et al., 2014)

Study parameters are exchangeable,
 $\theta_j \mid \mu, \tau^2 \sim N(\mu, \tau^2)$; the robust version adds a vague component that suppresses borrowing under conflict.

Commensurate prior (Hobbs et al., 2011, 2012)

$\theta \mid \theta_0 \sim N(\theta_0, 1/\tau)$ with a spike-and-slab hyperprior on the commensurability τ .

Power prior (Ibrahim and Chen, 2000)

Historical likelihood raised to a_0 .

Common feature

One heterogeneity or discount scale applies to the whole population, without depending on the patient's covariates. With a single historical study the three forms nearly coincide (Röver and Friede, 2025).

We allow borrowing to vary across covariate regions learned from the data.

The trial-control mean adds a discrepancy to the external mean

RCT controls ($D_i = 1$) and RWD controls ($D_i = 2$), covariates $\mathbf{x}_i$, outcome y_i :

$$y_i = f(\mathbf{x}_i) + \mathbb{I}\{D_i = 1\} g(\mathbf{x}_i) + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma_{D_i}^2)$$

f : external-control response mean

Standard BART, $H_f = 50$ trees, fitted to both sources. The RWD follow f .

g : discrepancy ensemble

Small and shallow, $H_g = 5$ trees, split prior $(0.5, 3)$, informed by the RCT controls only. The RCT controls follow $f_1 = f + g$.

- The decomposition follows Bayesian causal forests (Hahn et al., 2020), with source membership replacing treatment.
- The discrepancy is $g = f_1 - f$. Shrinking it toward zero centers the RCT-control mean f_1 at the external-control mean f .

Each discrepancy leaf has a spike-and-slab prior

The zero-centered mixture describes each leaf contribution to $g = f_1 - f$:

$\theta_{h\ell}$ is the contribution of leaf (h, ℓ) to g :

$$\theta_{h\ell} \mid w, \tau_0^2, \tau_1^2 \stackrel{\text{iid}}{\sim} w \text{N}(0, \tau_0^2) + (1 - w) \text{N}(0, \tau_1^2).$$

Equivalently, $z_{h\ell} \mid w \stackrel{\text{iid}}{\sim} \text{Bernoulli}(w)$ selects a zero-mean normal with variance $\tau_{1-z_{h\ell}}^2$. We use $w \sim \text{Beta}(1, 1)$, $\tau_0^2 \sim \text{scaled-Inv-}\chi^2(\nu_0, s_0^2)$ truncated to $(0, \tau_1^2)$, and τ_1^2 fixed by the range rule.

Spike, $z = 1$: the commensurate component

Shrinks a discrepancy contribution toward zero, drawing the RCT response mean toward $f(x)$. The ESS calibration sets the leaf scale τ_0 .

Slab, $z = 0$: the vague component

A free local offset on the outcome scale, corresponding to a robust-MAP vague component within a leaf.

Special cases

$w = 1$: commensurate BART (C-BART), one learned scale. $w = 1, \tau_0^2 \rightarrow 0$: BART on the pooled controls (BART-CP). $w = 0$: a free source offset, the tree analogue of BART with the source indicator (BART-PP).

The induced prior is centered at $f(\mathbf{x})$

- At a fixed profile $\mathbf{x}$, each discrepancy tree contributes one leaf: $g(\mathbf{x}) = \sum_{h=1}^{H_g} \theta_{h, \ell_h(\mathbf{x})}$, where $\ell_h(\mathbf{x})$ is the reached leaf.
- Integrating the leaf indicators gives a mixture over the number m of reached spike leaves:

$$f_1(\mathbf{x}) \mid f, T_g, w, \tau_0^2, \tau_1^2 \sim \sum_{m=0}^{H_g} \binom{H_g}{m} w^m (1-w)^{H_g-m} \mathcal{N}(f(\mathbf{x}), m\tau_0^2 + (H_g - m)\tau_1^2).$$

All spike: $m = H_g$

Variance $H_g\tau_0^2$: the commensurate component shrinks $f_1(\mathbf{x})$ toward $f(\mathbf{x})$.

All slab: $m = 0$

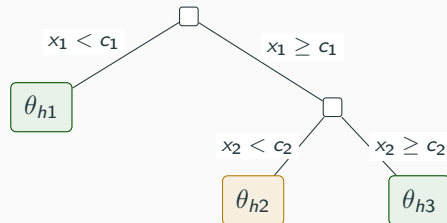
Variance $H_g\tau_1^2$: a larger departure is permitted, with the same center $f(\mathbf{x})$.

- T_g denotes the discrepancy partitions; every component is centered at the external mean $f(\mathbf{x})$.
- This is a prior for the latent control mean. Observation variance is separate.

A shared surface with local discrepancies

RWD controls: $f(x)$ **RCT controls:** $f(x) + g(x)$

One tree in g (illustrative splits)



Each profile reaches one leaf per tree.
Add those contributions: $g(x) = \sum_{h=1}^{H_g} \theta_{h, \ell_h(x)}$.

f : the external response mean

Fitted jointly to both control sources.

Spike: $z_{h\ell} = 1$

$\theta_{h\ell} \sim N(0, \tau_0^2)$

Draws the RCT mean toward $f(x)$.

Slab: $z_{h\ell} = 0$

$\theta_{h\ell} \sim N(0, \tau_1^2), \quad \tau_1^2 > \tau_0^2$

Allows departures from $f(x)$.

$z_{h\ell} \sim \text{Bernoulli}(w)$. Leaf indicators vary;
 w, τ_0^2, τ_1^2 are shared.

Separate ensembles allow different shrinkage priors

- Separate ensembles let us choose different partition and shrinkage priors for the external mean f and the discrepancy g .
- We use $H_f = 50$ trees for the outcome surface and $H_g = 5$ shallow trees for the discrepancy.
- At fixed leaf variances and partitions, fewer discrepancy trees reduce the total all-spike variance at a profile. The slab range rule also depends on H_g .

What the prior-density comparison establishes

Appendix G.6 minimizes over integer numbers of slab leaves and retains contributions from the remaining spike leaves. It compares joint prior densities at constrained modes. Posterior indicator probabilities also depend on the likelihood and partitions.

The simulations assess this specification. They do not establish that shared-leaf models cannot detect regional discrepancies.

Tree moves use exact leaf marginals

Conditionally on (τ_0^2, τ_1^2, w) the leaf prior is a finite Gaussian mixture, which gives closed-form factors for each tree move:

$$M_f = (1 + \lambda_f^2 A)^{-1/2} \exp\left\{\frac{\lambda_f^2 B^2}{2(1 + \lambda_f^2 A)}\right\}, \quad M_g = \frac{w \phi(\bar{y}_1; 0, \tau_0^2 + v_1) + (1 - w) \phi(\bar{y}_1; 0, \tau_1^2 + v_1)}{\phi(\bar{y}_1; 0, v_1)},$$

with A, B the precision-weighted counts and sums of both sources in an f leaf, and $\bar{y}_1, v_1 = \sigma_1^2/n_1$ the RCT residual mean and its variance in a g leaf.

- Grow, prune and change moves use the exact collapsed ratios, so the sampler targets the exact posterior without a surrogate likelihood.
- Leaf draws are conjugate; z is drawn first, then θ ; τ_0^2 and w update from the leaf means and states.
- For right-censored outcomes, log-normal AFT data augmentation leaves the other updates unchanged.
- One sweep costs a standard BART on $n_1 + n_2$ observations plus a five-tree ensemble on n_1 .

Prior effective sample size at the estimand

For $\mu = N^{-1} \sum_i f_1(x_i)$, V_μ^f is the RWD-only posterior variance. The working calibration is:

$$\text{ESS}_0(s_0^2) = \hat{\sigma}_1^2 \mathbb{E}_{\tau_0^2} \left[\frac{1}{V_\mu^f + c_g H_g \tau_0^2} \right].$$

- c_g averages squared leaf proportions before inversion; the scale law here is untruncated.
- A normal prior gives the ELIR variance ratio (Neuenschwander et al., 2020); this BART/scale-mixture summary is not exact marginal ELIR.
- Ceiling: $\hat{\sigma}_1^2 / V_\mu^f$, including BART regularization. Equality to scaled n_2 requires one fixed tree, matched proportions and a flat prior.
- The Pr-rule requires the target in 95% of blocks. A target at or above the whole-chain ceiling instead triggers a smallest-scale override.

Realized ESS diagnostic

With target leaf proportions $a_{h\ell}$, use $V_g = \sum_{h,\ell} a_{h\ell}^2 \{z_{h\ell} \tau_0^2 + (1 - z_{h\ell}) \tau_1^2\}$, then average $\sigma_1^2 / (V_\mu^f + V_g)$. Standardization can span several leaves per tree.

Conditional theoretical guarantees

Leaf results condition on f , the other discrepancy trees, partitions, scales and $0 < w < 1$. Write $v = \sigma_1^2/n$ for a leaf's RCT residual-mean variance.

Theorem 1: bounded local influence

Relative to the slab estimate, $\sup_{\bar{y}} |l(\bar{y})| \leq \sqrt{2v \max\{\log C, \frac{1}{2}\}}$, and the shift vanishes at a Gaussian rate under growing conflict. The C-BART and BART-CP leaf shifts are linear and unbounded.

Theorem 2: local detection and map consistency

The leaf spike probability decreases with $|\bar{y}|$. With fixed partitions, f fixed, a representable discrepancy and growing local counts, $\mathcal{B}_c(x) \rightarrow \mathbb{I}\{|d| < c\}$ for $|d| \neq c$. A fixed positive spike scale need not identify compatibility through its indicator.

Theorem 3: the working calibration

Under the untruncated law, ESS_0 is continuous, convex and strictly decreasing from its ceiling to zero. Truncation gives a positive lower limit. Grid-selection consistency requires the stated moment and separation conditions; ceiling overrides are separate.

The borrowing map

At each patient profile x , we report

$$\mathcal{B}_c(x) = P(|g(x)| < c \mid \text{data}),$$

the posterior probability that the local discrepancy is below a threshold c fixed in advance on the outcome scale, together with the posterior mean and interval of $g(x)$ and the pointwise ESS.

- $c = 0.5$ in the simulations, one third of the RCT residual SD for Gaussian outcomes.
- The all-spike probability $B(x)$ is kept as a diagnostic: its prior scale is w^{H_g} and at a fixed positive spike scale, it need not identify compatibility as RCT counts grow.
- In single-arm mode the data do not inform g , so the compatibility map is flat. The local ESS map describes variation across profiles.

Conditions for the map result

The map results require fixed g partitions, a representable discrepancy, growing local counts and f fixed; the result does not cover the posterior over partitions. The simulations show separation of shifted and compatible regions under the full posterior.

Simulation design

Common design

- $R = 500$ paired replicates
- $p = 10$ covariates, nonlinear surface
- 300 RWD controls
- Gaussian: RCT 200:100, $\sigma_1 = 1.5$, $\sigma_2 = 1.8$
- Survival: RCT 225:75, log-normal AFT, about a quarter censored
- Targets: absolute 50, 75, 100; fractional 0.25, 0.5, 0.9 of the ceiling

Five scenarios

- **Sc1** no confounding: source at random
- **Sc2** measured: source depends on (X_5, X_6)
- **Sc3** unmeasured: source and outcome depend on $\mathbf{U}$, observed as proxies at $\rho \in \{-0.5, 0, 0.5\}$
- **Sc4** regional shift: RWD surface raised by $\delta_2 \in \{0.5, 1, 2\}$ outside $\mathcal{R} = \{X_5 > 2\}$, half the population; secondary version on the null covariate X_7
- **Sc5** global shift: raised by δ_2 everywhere; borrowing should be suppressed

Illustrative leaf threshold: $d_{1/2} \approx 0.5$. Comparators: regression/BART pooling variants, PSCL, MAP, hierarchies, C-BART and LRC-BART ($w = 0$). CAHB (Jin et al., 2023): transferred analysis, first 200 paired Gaussian replicates; variants in the supplement.

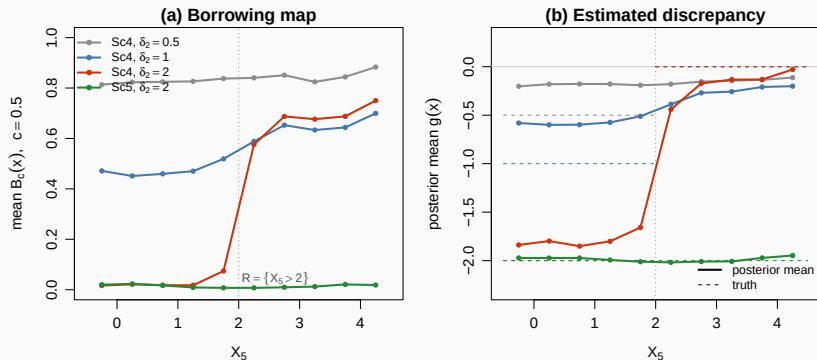
Gaussian outcomes: ATE bias, RMSE, coverage

Method	Sc1			Sc3, $\rho = 0$			Sc4, $\delta_2 = 1$			Sc4, $\delta_2 = 2$			Sc5, $\delta_2 = 2$		
	Bias	RMSE	Cov	Bias	RMSE	Cov	Bias	RMSE	Cov	Bias	RMSE	Cov	Bias	RMSE	Cov
LM-PP	-0.00	0.34	0.94	0.04	0.35	0.95	-0.01	0.34	0.95	-0.02	0.34	0.95	-0.04	0.34	0.94
BART-NP	0.00	0.22	0.91	-0.00	0.29	0.94	0.00	0.22	0.91	0.00	0.22	0.91	0.00	0.22	0.91
BART-CP	0.01	0.14	0.96	0.98	1.00	0.00	-0.34	0.37	0.35	-0.68	0.70	0.00	-1.39	1.40	0.00
BART-PP	0.01	0.21	0.94	0.03	0.27	0.94	-0.00	0.21	0.95	-0.01	0.21	0.96	-0.01	0.21	0.95
C-BART	0.01	0.15	0.95	0.23	0.38	0.87	-0.21	0.27	0.73	-0.23	0.39	0.66	-0.32	0.65	0.76
LRC-BART (100)	0.01	0.17	0.94	0.04	0.28	0.94	-0.04	0.22	0.89	0.00	0.20	0.92	-0.00	0.20	0.92

Agreement, global conflict and regional shifts

Agreement (Sc1): LRC-BART borrows, RMSE 0.172 against 0.205 for BART-PP; the pooling models have lower RMSE here. Global disagreement (Sc3, Sc5): it suppresses borrowing, with bias about 0.04 and 0.00, where C-BART and BART-CP have substantial bias. Regional shift: near-zero empirical bias at $\delta_2 = 2$; at $\delta_2 = 1$ it borrows across the boundary, bias -0.04 , coverage 0.89.

The borrowing map under a regional shift



LRC-BART at target 100, first 40 replicates

At $\delta_2 = 2$ the map is 0.02 outside $\mathcal{R}$ and 0.7 inside, and g is -1.8 and -0.15 against -2 and 0 ; the transition spans the unit around the boundary, with about four RCT controls per 0.1 of X_5 . At $\delta_2 = 1$ the estimated discrepancy captures part of the shift ($-0.6 / -0.25$). Under Sc5 the map is 0.01 everywhere.

Regional conflict remains difficult

Agreement and global conflict

Sc1: RMSE 0.172 versus 0.205 for BART-PP; realized ESS 53, prior ESS 82, ceiling 127. Under global conflict, realized ESS falls to 1.6 (Sc3) and 0.2 (Sc5). Sc3 bias is 0.012 above BART-PP (paired SE 0.001); the leaf theorem does not establish the cause of this difference.

Surface error under regional shifts

At shift 2 (1.3 residual SDs), compatible-region surface RMSE over all RCT profiles is 0.824 versus 0.785. On control profiles away from the boundary, it is 0.747 versus 0.723 (difference 0.024, SE 0.005, $R = 200$). At shift 1 (two thirds of a residual SD), bias is -0.04 , coverage 0.89, and RMSE 0.223 versus 0.207.

Regional ESS depends on the standardized mean

At shift 2, realized regional ESS is 9.5 in the compatible region and 0.44 outside ($R = 200$). Each uses its own standardized mean; they are not additive. The compatibility map reports a posterior probability.

The CAHB comparison assesses our transfer of kernel-local estimation; it does not establish a general method ranking. Sc2's ceiling falls from 127 to 23, so absolute targets are capped.

Method	Sc1, median ratio			Sc3 $\rho = 0$, median ratio			Sc3 $\rho = 0$, RMST ratio		
	Bias	RMSE	Pow	Bias	RMSE	Cov	Bias	RMSE	Cov
AFT-NP	0.18	0.67	0.12	0.23	0.83	0.94	-0.03	0.15	0.92
BART-NP	0.26	0.51	0.46	0.19	0.57	0.90	0.06	0.17	0.92
BART-CP	0.06	0.23	0.63	2.31	2.41	0.00	0.59	0.61	0.01
BART-PP	0.03	0.35	0.25	0.04	0.45	0.91	-0.01	0.14	0.92
C-BART	0.03	0.25	0.42	0.58	0.91	0.88	0.14	0.22	0.85
LRC-BART (100)	0.05	0.31	0.34	0.15	0.56	0.94	0.03	0.15	0.94

Median and RMST ratios

With 75 controls and censoring, the median ratio has a wide, skewed posterior. Its posterior mean has upward bias in Sc1. Borrowing reduces RMSE (Sc1: RMSE 0.312 against 0.345, power 0.34 against 0.25 for BART-PP). Under unmeasured confounding LRC-BART suppresses borrowing (realized ESS 2.3) and has 0.11 more median-ratio bias than BART-PP, alongside its wider posterior and the nonlinear ratio transformation. On the RMST ratio the biases are 0.03 and -0.005 .

Single-arm trials cannot learn a control discrepancy

Trial contributes treated subjects only ($n_T = 200$ Gaussian; 30 or 200 survival); the 300 RWD controls are the sole control source; LRC-BART in single-arm mode with $w = 1$.

No confounding

BART-CP and LRC-BART have Gaussian RMSE 0.16; the survival RMST ratio at $n_T = 30$ has RMSE 0.16 to 0.19 with coverage 0.92 to 0.97.

Measured confounding

Every method is biased: Gaussian ATE bias 1.68 (LM-CP), 0.41 (BART-CP), 0.42 (LRC-BART); there are no internal controls to identify a discrepancy, and RWD overlap is limited.

- The single-arm ceiling is 27 to 34 under Sc2, so an absolute target of 100 is capped and LRC-BART approximates pooling with $w = 1$; the fractional target still widens the interval (coverage 0.82 to 0.99).
- With $w < 1$, slab variance $(1 - w)H_g\tau_1^2$ adds uncertainty: SD 1.4 for the ATE and 10 to 34 for the median ratio. Zero-mean g preserves linear log-time means, but survival-ratio means also change. Primary analysis: $w = 1$.
- Hierarchical regressions: ATE SD 8 under the stated hyperprior.

EloKRd phase II trial

- $n = 30$, all received EloKRd
- 13 PFS events; 10 deaths
- 50% high-risk cytogenetics; 30% ASCT

UCMM registry control

- $n = 253$ after cohort restrictions
- Standard induction regimens
- Five baseline covariates plus ASCT after induction

Harmonization and the scope of the comparison

Inconsistent non-Hispanic indicator coding changes extrapolation. Harmonization raises the working ESS ceilings from 12.6/14.6 to 63.7/36.7 (PFS/OS). ASCT is recorded after induction, so this adjustment supports a descriptive comparison, not a total causal treatment effect.

Estimand: standardized 5-year RMST ratio (neither trial median is reached). Kaplan–Meier: 1.015 for PFS, 0.90 for OS.

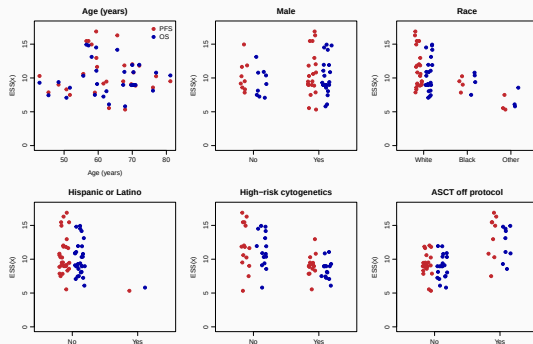
EloKRd results: 5-year RMST ratio, posterior mean and 95% interval

Harmonized coding	PFS	OS
Kaplan–Meier, unadjusted	1.01 (0.84, 1.22)	0.90 (0.79, 1.03)
Complete-pooling AFT	0.87 (0.69, 1.06)	0.80 (0.65, 0.94)
Hierarchical AFT	1.23 (0.74, 1.90)	1.21 (0.77, 1.70)
AFT-BART on UCMM, no discrepancy	0.97 (0.79, 1.16)	0.90 (0.78, 1.02)
LRC-BART, $N_{\text{target}} = 30$	0.97 (0.77, 1.23)	0.90 (0.77, 1.04)
LRC-BART, $N_{\text{target}} = 15$	0.99 (0.75, 1.40)	0.91 (0.76, 1.11)
LRC-BART, $0.9\times$ ceiling	0.97 (0.79, 1.15)	0.90 (0.77, 1.04)
<i>Earlier analysis, original coding: MAP-AFT-BART, 30</i>	1.08 (0.66, 16.4)	0.97 (0.74, 4.77)

Descriptive comparison conditional on recorded profiles

At target 30, $P(\text{ratio} > 1)$ is 0.37 (PFS) and 0.08 (OS); both nonparametric intervals include one. Lower targets widen intervals. The fitted PFS distribution has a 99.9th percentile of 1.6 over 8,000 draws, but the earlier sampler is unavailable, so we cannot isolate the cause of the tail difference.

The local ESS map at the 30 trial profiles



Registry support at the trial profiles

$ESS(x) = \hat{\sigma}_1^2 / \{V_f(x) + H_g \bar{\tau}_0^2\}$ ranges from 5 to 17 for PFS and 6 to 15 for OS: highest for White/ASCT/standard-risk profiles; lowest for the one Hispanic patient, Other/Unknown race and high-risk profiles without ASCT. $\bar{\tau}_0^2$ is the fitted mean scale. Target 30 concerns the standardized log-time mean; local values are not additive.

Main findings

1. The RCT control mean is centered on the external mean through a spike-and-slab discrepancy ensemble.
2. Closed-form leaf marginals support posterior-invariant tree updates; this does not establish finite-chain mixing.
3. Variance-based ESS calibration concerns the standardized control mean. The calculation specifies its ceiling and approximation convention.
4. Conditional leaf influence and map results require the stated fixed-surface and partition assumptions.
5. Gaussian Sc1 RMSE is about 16% lower than BART-PP's. Regional conflict produces informative compatibility summaries, but also surface losses and excessive borrowing at moderate shifts.

Limitations and next steps

- Full-posterior map consistency over learned partitions is unproved; the Sc4 surface loss persists away from the boundary.
- Calibration uses a working precision summary. Exact marginal ELIR would require a different calculation and new calibration analyses.
- Multiple external sources require joint calibration. Additive ceilings have not been established.
- A single-arm study cannot learn the control discrepancy. The application remains a descriptive comparison with post-induction ASCT adjustment.
- Confirmatory use requires operating-characteristic evaluation, including Type I error and power, and a pre-specified role for the map.

How to interpret the borrowing summaries

The compatibility map, prior pointwise ESS and realized regional ESS answer different questions. These summaries do not give an observed count of external patients used at each profile.

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